

Supplementary Material: Boosting Multimodal Remote Sensing Classification via Intra-modal Discriminability Modulation

1 Comparison of Ambiguity Scoring Functions

To justify the choice of the top-2 ratio in Eq. (5) of the main paper, we replace the ambiguity score a_i^m with three alternative uncertainty measures while keeping all other components of IDM unchanged. The candidates are:

- **Max-Prob:** $a_i^m = p_{i,(1)}^m$, the maximum softmax probability;
- **Norm-Entropy:** $a_i^m = 1 - H(\mathbf{p}_i^m) / \log C$, where $H(\cdot)$ denotes Shannon entropy, so that larger values still indicate lower ambiguity;
- **Gini:** $a_i^m = \sum_c (p_{i,c}^m)^2$, the sum of squared probabilities (higher means more concentrated).

All variants share the same self-calibrated scale strategy (Eqs. (12)–(13) of the main paper) and identical warm-up epochs. Results on Houston (HS/DSM) and Augsburg (HS/SAR) with ResNet18 are reported in Table S1.

Table S1: Comparison of ambiguity scoring functions.

Scoring fn.	Houston (HS/DSM)			Augsburg (HS/SAR)		
	OA(%)	AA(%)	Ka.(%)	OA(%)	AA(%)	Ka.(%)
w/o $\mathcal{L}_{\text{reg}}$	82.36	82.13	80.89	84.93	50.02	77.85
Max-Prob	83.73	84.18	82.37	86.17	51.52	79.70
Norm-Entropy	84.06	84.27	82.73	86.44	50.43	79.96
Gini	84.14	84.37	82.82	86.73	50.88	80.43
Top-2 (ours)	84.75	85.51	83.48	88.00	56.19	82.43

All four scoring functions substantially outperform the unregularized baseline. Among them, the top-2 ratio achieves the best overall performance. Max-Prob generally yields the lowest OA improvements because the maximum softmax probability saturates near 1 during later training, compressing the effective modulation range. Norm-Entropy and Gini incorporate all C classes, but the near-zero tail entries contribute marginal information while introducing extra sensitivity to logit scale fluctuations. The top-2 ratio avoids both issues, providing a stable and discriminative modulation signal throughout training.

2 Noise Generation Details

We design physically motivated noise models for each sensor type. The corruption level $\epsilon \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$ controls the perturbation magnitude relative to the signal’s standard deviation σ_x . All compared methods are evaluated on identical corrupted inputs.

HS: Additive Gaussian Noise. Independent per-channel noise simulates photon shot noise that disrupts spectral consistency:

$$\tilde{\mathbf{x}}^{\text{HS}} = \mathbf{x}^{\text{HS}} + \epsilon \cdot \sigma_x \cdot \mathbf{n}, \quad \mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{S1})$$

SAR: Multiplicative Speckle Noise. An energy-preserving log-normal model simulates coherent speckle:

$$\tilde{\mathbf{x}}^{\text{SAR}} = \mathbf{x}^{\text{SAR}} \odot \exp(\epsilon \cdot \mathbf{n} - \frac{1}{2}\epsilon^2), \quad \mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (\text{S2})$$

where the $-\frac{1}{2}\epsilon^2$ term ensures $\mathbb{E}[\exp(\epsilon \cdot \mathbf{n} - \frac{1}{2}\epsilon^2)] = 1$ for $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

DSM: Sparse Impulse Noise. Large-magnitude outliers on approximately 30% of pixels simulate anomalous LiDAR returns:

$$\tilde{\mathbf{x}}^{\text{DSM}} = \mathbf{x}^{\text{DSM}} + \mathbf{m} \odot (5\epsilon \cdot \sigma_x \cdot \mathbf{n}), \quad m_{ij} \sim \text{Bernoulli}(0.3). \quad (\text{S3})$$

3 Detailed Noise Robustness Results

Table S2 reports the full OA (%) values corresponding to Fig. 5 in the main paper. On the Augsburg dataset under joint corruption at $\epsilon = 1.0$, IDM outperforms the Baseline by +35.69 percentage points (59.31% vs. 23.62%).

4 t-SNE Visualization

Fig. S1 visualizes the learned feature distributions via t-SNE under increasing noise on both datasets. The noise models follow Sec. 2, with one exception: on Houston (HS/DSM), the HS corruption is intensified to signal-independent Gaussian noise ($\tilde{\mathbf{x}} = \mathbf{x} + \varphi\epsilon$, where φ is a fixed scalar controlling the absolute perturbation magnitude) to stress-test the fusion under dominant-modality failure, as Houston is highly HS-dependent. This intensified protocol is marked by the dashed border in Fig. S1(a). For both the class-pair views and the all-class projections, 500 samples per class are uniformly sampled to ensure balanced representation.

In the class-pair views (left three groups), each group isolates two classes that the Baseline model most frequently confuses, to examine their separation under noise. As noise increases, the Baseline clusters increasingly overlap, while IDM maintains tighter separation. This contrast is especially visible in Houston at higher noise levels, where the Baseline red and blue points become nearly inseparable, yet IDM still preserves distinct cluster boundaries. The all-class projections (rightmost group) show a consistent pattern: under clean input, both methods produce reasonable cluster structure, but as noise increases, the Baseline clusters fragment and merge across classes, whereas +IDM retains more compact per-class groupings. On Augsburg (Fig. S1(b)), the degradation is more gradual due to the channel-adaptive HS noise, but the relative advantage of +IDM remains visible across all noise levels.

5 Parameter Sensitivity Analysis

We jointly tune the scaling factor c and the regularization weight λ on the Houston and Augsburg datasets; results are shown in Fig. S2. IDM maintains competitive performance over a relatively broad range of hyperparameter values. As c increases, the performance peak tends to shift toward smaller λ , consistent with the wider modulation range induced by a larger c . Only under relatively extreme settings does performance degrade noticeably. Compared with Augsburg (HS/SAR), Houston (HS/DSM) exhibits noticeably larger fluctuations across hyperparameter settings, which may be attributed to the stronger modality disparity in this dataset, making the regularization more prone to over-strengthening when both c and λ are large.

Table S2: OA (%) under noise perturbation on Houston (HS/DSM) and Augsburg (HS/SAR). Best results per setting are bolded.

Corruption	Method	Houston (HS/DSM)						Augsburg (HS/SAR)					
		0.0	0.2	0.4	0.6	0.8	1.0	0.0	0.2	0.4	0.6	0.8	1.0
HS	Baseline	82.36	81.29	71.27	54.04	41.32	34.09	84.93	85.09	84.90	82.41	74.67	59.72
	OGM-GE	83.41	82.73	70.32	51.46	38.72	34.16	84.58	84.69	83.98	79.42	68.23	49.73
	DRBML	83.15	82.34	78.74	67.26	50.79	39.07	84.89	84.91	84.45	80.42	67.10	41.48
	ARL	83.28	82.13	76.45	65.89	52.82	41.99	85.08	85.19	84.63	79.72	65.94	42.19
	IDM	84.75	84.50	82.36	74.75	63.06	52.43	88.00	87.87	87.06	83.95	77.77	70.84
DSM/ SAR	Baseline	82.36	82.04	78.75	73.67	68.48	63.85	84.93	73.96	69.41	64.24	60.10	57.12
	OGM-GE	83.41	82.92	80.12	76.41	71.76	66.62	84.58	74.51	68.76	62.19	56.66	52.52
	DRBML	83.15	82.65	80.85	78.10	74.24	70.08	84.89	72.92	66.86	60.90	56.45	53.11
	ARL	83.28	82.66	79.58	75.15	69.30	63.91	85.08	76.24	71.48	65.78	60.81	56.86
	IDM	84.75	84.26	81.96	78.54	75.08	72.06	88.00	81.22	78.62	76.67	75.14	73.36
Both	Baseline	82.36	80.15	66.39	45.86	33.43	26.52	84.93	74.45	68.07	56.61	42.52	23.62
	OGM-GE	83.41	81.98	66.23	42.87	28.45	20.80	84.58	74.15	64.76	50.91	34.59	16.15
	DRBML	83.15	81.91	76.30	58.63	39.03	25.55	84.89	72.63	64.07	51.43	30.83	11.12
	ARL	83.28	81.50	73.02	55.84	39.46	28.95	85.08	75.57	65.67	51.77	33.29	13.08
	IDM	84.75	84.18	79.22	65.84	48.74	35.51	88.00	81.04	78.19	73.39	66.30	59.31

The insets in Fig. S2 display the automatically calibrated σ^m values when $c = 1$. The relative ordering of σ^m across modalities is consistent with the discriminability gap reported in prior work, suggesting that the proposed calibration mechanism adapts the modulation range to different modality combinations in a reasonable manner.

6 Seed Sensitivity and Statistical Significance

The main-paper results use a single fixed seed for reproducibility. To verify that the reported gains are systematic rather than seed-specific, we repeat the experiments on Augsburg (HS/DSM), the dataset with the smallest average improvement, using five strong backbones with five random seeds each. All factors are held identical with and without IDM, so the two settings differ only in whether IDM is applied. Table S3 reports the mean and standard deviation over seeds.

IDM improves the mean of all three metrics on every backbone except AA on DSymFuser, where the two settings are statistically indistinguishable. A one-sided paired Wilcoxon signed-rank test over the 25 paired results confirms that the improvement is significant on all three metrics. IDM also reduces the across-seed standard deviation in most entries, indicating that the modulation additionally stabilizes training rather than merely shifting the mean.

7 Computational Cost

The additional cost of IDM stems mainly from constructing the unimodal proxy outputs, i.e., M extra forward passes through the fusion module and classifier (M = number of modalities), plus logit-level operations (ambiguity scoring, Gaussian modulation, and KL). Let B be the batch size, C the number of classes, and C_{fus} the cost of one fusion-and-classification pass. The total increment is

$$\Delta C \approx M C_{\text{fus}} + O(MBC), \quad (\text{S4})$$

where the second term comes from the logit-level operations. Since M , B , and C are fixed under a given setup, the overhead depends

mainly on C_{fus} and its share of the total forward cost, i.e., on the fusion-module design rather than on IDM itself.

Fig. S3(a) reports the measured training-time overhead across backbones. For models with lightweight fusion modules the increase is marginal, e.g., +3.8%, +4.1%, and +0.4% for ResNet18, DSymFuser, and DAHGM respectively, whereas models with computation-heavy fusion such as S2Mamba show a larger increase, consistent with the analysis above.

Because the proxy passes are executed sequentially rather than in parallel, peak GPU memory is identical for the Baseline and +IDM within profiling resolution: 1013 MB (ResNet18), 835 MB (CO-CNN), 1061 MB (MFT), 3769 MB (ExViT), 2149 MB (DSymFuser), 2255 MB (S2Mamba), and 2671 MB (DAHGM). As a training-side method, IDM introduces no additional inference cost: at test time the model uses the standard fusion branch only.

8 Extension to More Than Two Modalities

IDM extends beyond two modalities without any structural change, since every modality is treated identically: the ambiguity score, the Gaussian modulation, and the scale σ^m are all defined per modality, and Eq. (14) of the main paper already sums $\mathcal{L}_{\text{reg}}^m$ over all M modalities. We verify this on the three-modality Augsburg (HS/SAR/DSM) setting, keeping all training hyperparameters identical to Sec. 4.1 of the main paper. As shown in Fig. S3(b), IDM improves over both the joint-training baseline and the competing modality-level modulation methods on all metrics, indicating that the per-sample allocation remains effective as the number of modalities grows.

9 Alternative Kernels and Fused-Target Reliability

9.1 Choice of the Modulation Kernel

The behaviour of IDM does not depend on the Gaussian specifically, but on the modulation profile derived in Sec. 3.3 of the main paper. We choose the Gaussian because it gives the simplest analytical

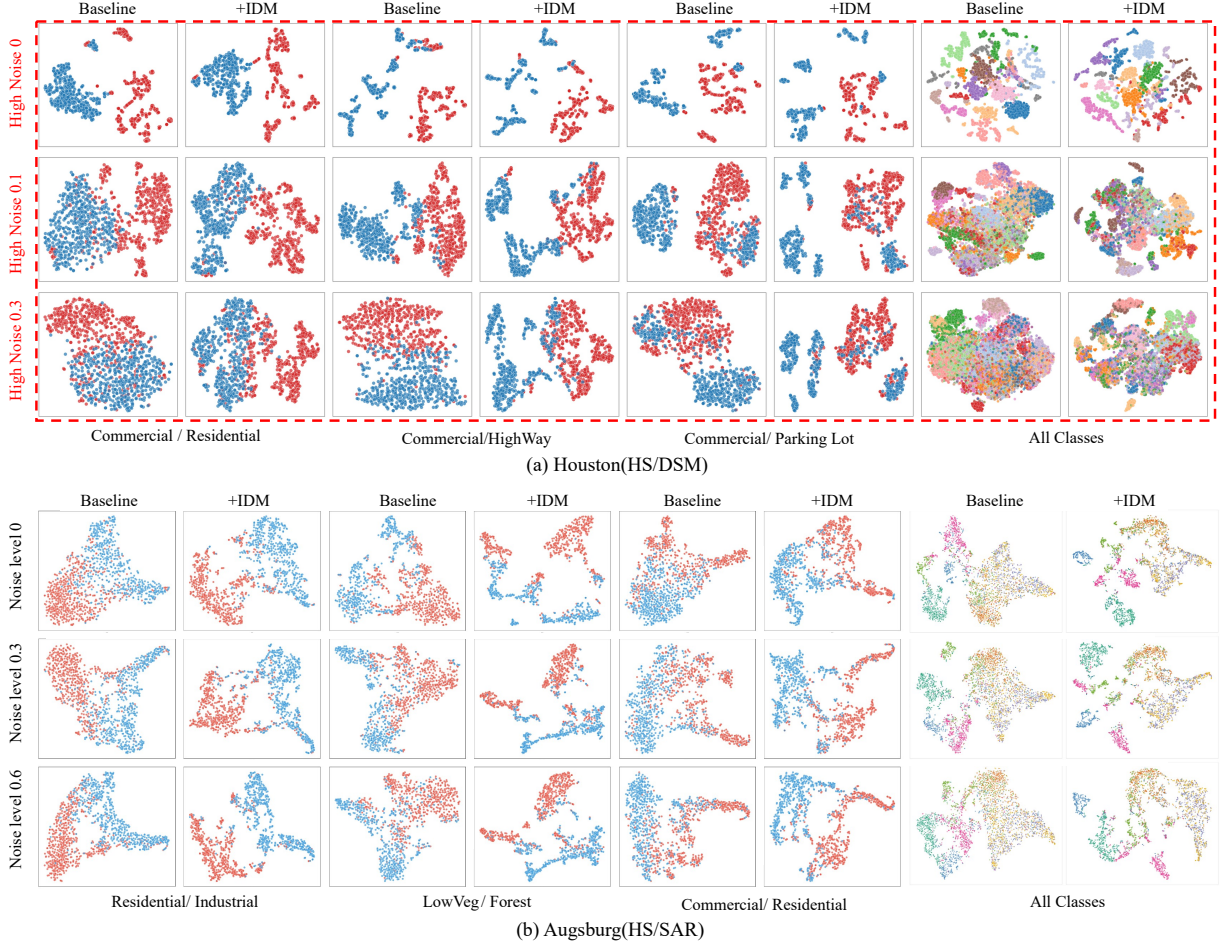


Figure S1: t-SNE visualization of learned features under increasing noise. Left three groups: class-pair scatter plots for confusable class pairs selected by Baseline confusion frequency. Rightmost group: all-class projection. Each column pair compares the Baseline and IDM under the same noise level. The dashed border in (a) indicates the intensified HS noise protocol used on Houston.

Table S3: Seed sensitivity on Augsburg (HS/DSM), reported as Baseline $\rightarrow$ +IDM (mean $_{\pm\text{std}}$ over five seeds).

Backbone	OA (%)	AA (%)	Kappa (%)
CO-CNN	89.80 $\pm$ 1.12 $\rightarrow$ 90.29 $\pm$ 0.55	63.70 $\pm$ 1.70 $\rightarrow$ 64.30 $\pm$ 1.29	85.22 $\pm$ 1.54 $\rightarrow$ 85.87 $\pm$ 0.76
MFT	85.97 $\pm$ 1.34 $\rightarrow$ 86.75 $\pm$ 1.13	57.21 $\pm$ 2.41 $\rightarrow$ 59.82 $\pm$ 1.10	79.98 $\pm$ 1.82 $\rightarrow$ 81.08 $\pm$ 1.55
DSymFuser	89.21 $\pm$ 0.88 $\rightarrow$ 90.10 $\pm$ 0.72	67.47 $\pm$ 3.08 $\rightarrow$ 67.43 $\pm$ 3.04	84.57 $\pm$ 1.24 $\rightarrow$ 85.88 $\pm$ 1.01
S2Mamba	90.73 $\pm$ 0.82 $\rightarrow$ 91.03 $\pm$ 0.73	64.93 $\pm$ 1.97 $\rightarrow$ 65.71 $\pm$ 2.06	86.59 $\pm$ 1.30 $\rightarrow$ 87.14 $\pm$ 0.99
DAHGM	90.24 $\pm$ 0.44 $\rightarrow$ 90.61 $\pm$ 0.19	66.47 $\pm$ 2.42 $\rightarrow$ 67.16 $\pm$ 2.57	86.01 $\pm$ 0.60 $\rightarrow$ 86.53 $\pm$ 0.30
Mean Δ	+0.57***	+0.93**	+0.82***

One-sided paired Wilcoxon signed-rank test over 25 paired results (5 backbones $\times$ 5 seeds). ** $p < .01$, *** $p < .001$.

control over the balance between the two signals: since

$$\frac{|\mathcal{G}'(a^m)|}{\mathcal{G}(a^m)} = \frac{a^m}{(\sigma^m)^2}, \quad (\text{S5})$$

the enhancement drive strengthens linearly relative to the guidance weight as a^m increases, i.e., as the modality becomes less ambiguous, with the slope set by the calibrated scale σ^m . By contrast, a Laplacian

kernel yields a constant ratio that cannot adapt this balance with a^m , while an inverse-quadratic kernel yields a nonlinear ratio that is less directly controlled by σ^m .

Table S4 substitutes the Gaussian with these two alternatives, keeping all other components unchanged. Both alternatives still improve over the unregularized baseline, confirming that the gain

Table S4: Alternative modulation kernels and the oracle-gated KL variant.

Variant	Houston (HS/DSM)			Augsburg (HS/SAR)		
	OA	AA	Ka.	OA	AA	Ka.
IDM (full)	84.75	85.51	83.48	88.00	56.19	82.43
$\mathcal{G} \rightarrow$ Laplacian	84.61	84.69	83.30	86.76	52.01	80.98
$\mathcal{G} \rightarrow$ inv.-quadratic	83.51	84.12	82.14	86.67	51.66	80.52
KL: oracle-gated	84.44	85.45	83.15	88.33	53.02	82.79

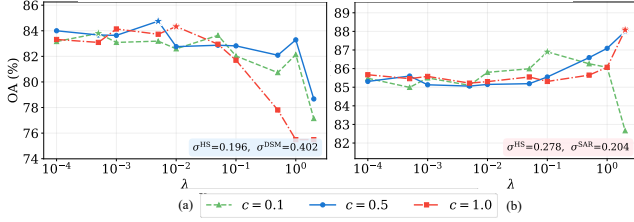


Figure S2: Parameter sensitivity on (a) Houston (HS/DSM) and (b) Augsburg (HS/SAR). Each curve fixes c and sweeps λ ; insets show the calibrated σ^m .

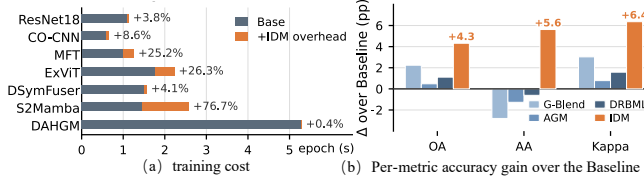


Figure S3: (a) Training-time overhead of IDM across backbones. (b) Results on the three-modality Augsburg (HS/SAR/DSM) setting, comparing IDM with the joint-training baseline and modality-level modulation methods.

comes from the modulation profile rather than the specific kernel, but the Gaussian performs best overall, with the margin most pronounced on Augsburg.

9.2 Effect of Incorrect Fused Targets

The KL term in Eq. (9) of the main paper treats the fused prediction as a soft target, which raises the question of what happens when that target is itself wrong. Three factors limit the impact. First, even a soft target with an incorrect argmax tends to carry useful class-affinity information, and the task cross-entropy on $\mathbf{p}_i^f$ acts to correct rather than reinforce such a target [1]. Second, the KL term is normalized by $1/\log C$ and carries a stop-gradient on $\mathbf{p}_i^f$, which bounds its influence relative to the task loss. Third, the guidance weight $\mathcal{G}(a_i^m)$ is largest exactly where the modality is most ambiguous, so confident modalities are barely affected.

To examine this empirically, we add an oracle-gated variant that applies the KL term only on samples where $\mathbf{p}_i^f$ is already correct. As shown in the last row of Table S4, this variant performs comparably to the full IDM, suggesting that the influence of occasionally incorrect fused targets is limited in practice.

- [1] Guangda Ji and Zhanxing Zhu. 2020. Knowledge distillation in wide neural networks: Risk bound, data efficiency and imperfect teacher. *Advances in Neural Information Processing Systems* 33 (2020), 20823–20833.